

Magic of Exponentials

🌾 Module 1: The ancient Indian story of the Chaturanga

🌾 1.1 The Legend of Chaturanga

In ancient India, the inventor of chess (Chaturanga) presented his game to the ruling king. As a reward, the inventor requested a seemingly trivial prize:

- **1 grain of rice** on the first square of the board.
- **2 grains** on the second square, **4 grains** on the third, **8 grains** on the fourth, and so on.
- The payout must double continuously for each of the **64 squares** on the board.
- The king immediately accepted, how much rice did the king owe to the inventor? Can you guess at the answer? This picture should give you a clue!



In this session we will calculate this from first principles which will show the real nature of Exponential functions following the

10 steps below:

1. First let us draw a 4x4 chess board with 16 squares:

sq #	sq #	sq #	sq #
1	2	3	4
5	6	7	8
??	??	??	??
??	??	??	??

...

2. Now, let us put the number of rice in each square:

rice qty	rice qty	rice qty	rice qty
1	2	4	8
16	??	??	??
??	??	??	??
??	??	??	??

...

3. This can be modeled as a series exponential numbers as follows:

[note that $2^0 = 1$ or $(any\ number)^0 = 1$]

2^0	2^1	2^2	2^3
2^4	??	??	??
??	??	??	??
??	??	??	2^{15}

why are we missing 2^{16} above ?

4. Now, let us create a Series with 16 terms for 16 squares and sum all of the terms:

$$S_{16} = 1 + 2 + 4 + 8 + 16 + 32 + 64 + \dots + 32,768$$

5. You can also write the 16-term series using exponentials as (hints: $2^0 = 1$ and $2^1 = 2$):

$$S_{16} = 2^0 + 2^1 + 2^2 + 2^3 + 2^4 + \dots + 2^{14} + 2^{15} \text{ why are we missing the } 2^{16} \text{ term?}$$

6. Can you write out the full Series for a 8x8 square chessboard in powers of 2? (you will need 64 terms):

$$\textcircled{1} S_{64} = 2^0 + 2^1 + 2^2 + 2^3 + 2^4 + \dots + 2^{62} + 2^{63}$$

7. Multiply equation $\textcircled{1}$ above by 2 on both sides of the = sign:

$$\textcircled{2} 2 * S_{64} = 2 * 2^0 + 2 * 2^1 + 2 * 2^2 + 2 * 2^3 + 2 * 2^4 + \dots + 2 * 2^{62} + 2 * 2^{63}$$

8. Simplify equation $\textcircled{2}$ above to get (hints: $2 * 2^0 = 2^{1+0} = 2^1$ and $2 * 2^3 = 2^{1+3} = 2^4$):

$$\textcircled{3} 2 * S_{64} = 2^1 + 2^2 + 2^3 + 2^4 + 2^5 + \dots + 2^{63} + 2^{64}$$

9. Subtract equation $\textcircled{1}$ from equation $\textcircled{3}$ to get $\textcircled{3} - \textcircled{1}$ as follows (shifted terms in equation $\textcircled{3}$ for lining up):

$\textcircled{3} 2 * S_{64} =$	(shift) +	$2^1 +$	$2^2 +$	$2^3 +$	$2^4 +$	$2^5 +$	...	$2^{63} +$	2^{64}	
$\textcircled{1} S_{64} =$	$2^0 +$	$2^1 +$	$2^2 +$	$2^3 +$	$2^4 +$	...	$2^{62} +$	2^{63}		
$\textcircled{4} S_{64} =$	$-1 +$	$0 +$							2^{64}	

10. Simplify equation $\textcircled{4}$ above to get $\textcircled{5}$ which represents the number of rice kernels in square 64 of the chess board.

$$\textcircled{5} S_{64} = 2^{64} - 1$$

Computing that that in your calculator will give you a HUMUNGOUS number - 18 followed by another 18 decimals or 18.5 quntillion rice kernels!!

$$\textit{Total Amount of Rice} = (2^{64} - 1) = 18,446,744,073,709,551,615 \text{ kernels}$$

Just how much rice is that when compared to the entire annual rice production of India? How many years will it take to

produce that much rice?

Here is some pertinent information for you to calculate this

1. Average Weight of a Single Rice Grain ≈ 0.025 grams
2. Indian Annual Rice Production = 150 million metric tonnes
3. There are 1000 grams in a kg
4. There are 1000 kg in a metric tonne

Let us work on this together in class ...



1.2 Homework Question/Problem

- Would you rather receive **1 Lakh rupees (₹100,000) every day for a whole month**, or **10 paise (₹0.10) that doubles each day for a month?**
- Hint: Convert 10 paise to 0.1 rupees and do exactly the same process we did in calculating # of rice grains.
- Another Hint: Same formula ($S_{64} = 2^{64} - 1$) will work, except you have to change 64 to another number
- Examine plot of results below and write down your thoughts about it.
- We will discuss all of this the following week!